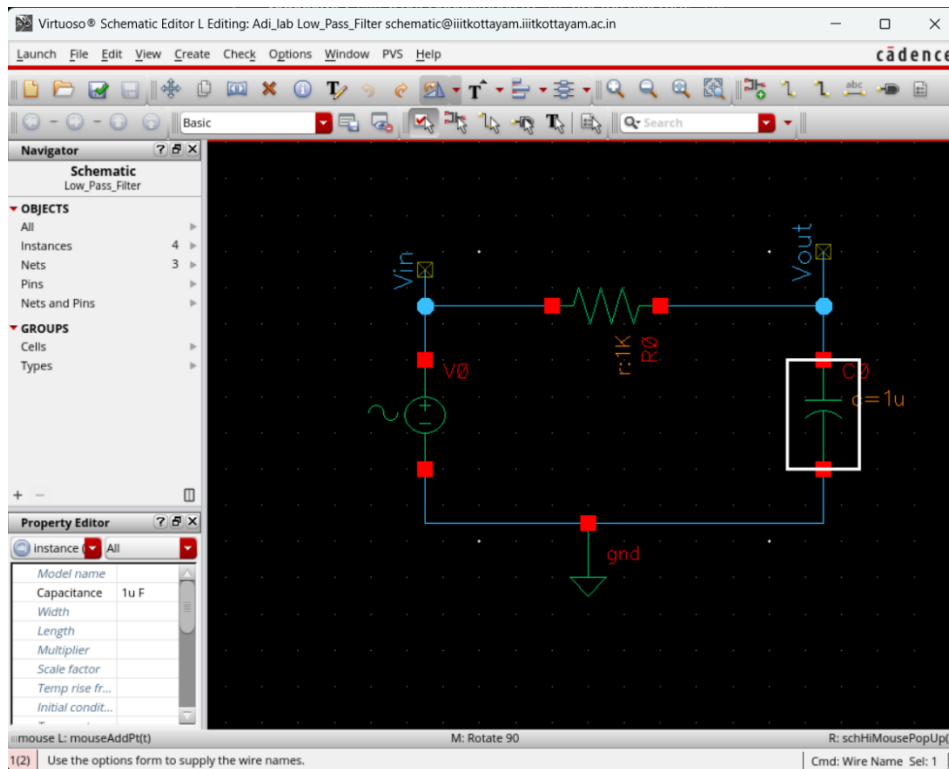


Low Pass Filter

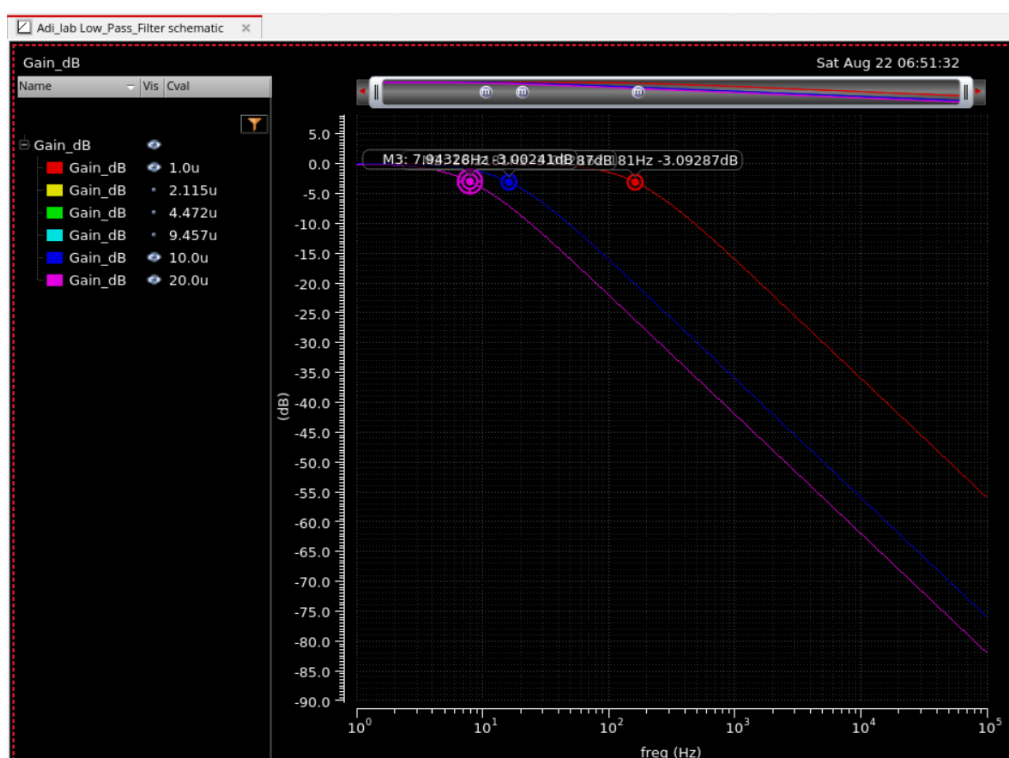
Adithyan Anilkumar

2025BEC0014

1. Schematic



2. Gain plot in decibels



The -3dB Bandwidth of the filter is inversely proportional to the capacitance value

$$f_c = 1/(2\pi RC)$$

The simulated –3 dB cutoff frequencies obtained from the Cadence AC sweep approximately match the theoretical cut-off frequency values

Capacitance	Theoretical value	Simulated value
1 μ F	159.15 Hz	158.9 Hz
10 μ F	15.91 Hz	16 Hz
20 μ F	7.96 Hz	7.94 Hz